

Mathematical Expectation : / Expected value of a r.v.

Defⁿ: If X is a discrete random variable having a PMF $f(x)$, then the expected value of X or the mathematical expectation of X is denoted with $E(X)$ and is defined by as

$$E(X) = \sum_x x f(x)$$

[In other words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes that value.]

On the other hand, if X is a continuous random variable having a PDF $f(x)$, then the expected value of X is defined by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Expected Value = Mean

$$E(X) = \text{Mean}$$

Theorem :

Let, X be a discrete random variable with probability function $f(x)$ and c be a constant. Then $E(c) = c$.

Proof : By definition,

$$E(c) = \sum_x c f(x)$$

$$= c \sum_x f(x)$$

$$= c \cdot 1 \quad \left[\because \sum_x f(x) = 1 \text{ for PMF} \right]$$

$$\therefore E(c) = c$$

(Proved)

Defⁿ :

Let, X be a random variable with probability distribution $f(x)$, then the expected value of the function $w(X)$ of the random variable X is

$$E[w(X)] = \sum_x w(x) f(x); \text{ if } X \text{ is discrete.}$$

$$\& E[w(X)] = \int_{-\infty}^{\infty} w(x) f(x) dx; \text{ if } X \text{ is continuous.}$$

Variance: Let, X be a random variable with finite mean $\mu = E(X)$, then the variance of X is denoted by $V(X)$, and is defined as

$$V(X) = E(X - \mu)^2$$

$$\text{or, } V(X) = E[(X - E(X))^2]$$

Standard Deviation: The positive square root of the variance is known as the standard deviation

$$\text{i.e. } \sigma = \sqrt{V(X)} = \sqrt{E(X - \mu)^2}$$

So, we can write,

$$V(X) = \sigma^2$$

Theorem: Let X be a discrete random variable with probability mass function $f(x)$; then

$$V(X) = \sigma^2 = E(X - \mu)^2 = E(X^2) - \mu^2$$

Proof: By definition.

$$V(X) = \sigma^2 = E(X - \mu)^2; \text{ where } \mu = E(X)$$

$$\Rightarrow \sigma^2 = E(X^2 - 2X\mu + \mu^2)$$

$$\begin{aligned}
 \Rightarrow \sigma^2 &= E(X^2) - 2\mu E(X) + E(\mu^2) \\
 &= E(X^2) - 2\mu \cdot \mu + \mu^2 \\
 &= E(X^2) - 2\mu^2 + \mu^2 \\
 \therefore \sigma^2 &= E(X^2) - \mu^2
 \end{aligned}$$

where,

$$E(X^2) = \sum_x x^2 f(x)$$

Th^m

The expected value of the sum of two random variables X and Y is the sum of the expected values of the variables, symbolically,

$$E[X+Y] = E(X) + E(Y)$$

Th^m

Let X be a random variable with a finite mean. Then for any numerical constants a and b ,

$$E(aX+b) = aE(X) + b$$

Th^m :

The expected value of the two random variables X and Y is equal to the product of their expected values, only when the variables are independent, i.e.,

$$E(XY) = E(X) \cdot E(Y)$$

In other words, the expected value of the product of two random variables is equal to the product of their expectations.

Moment Generating Function (MGF) :

Let, X be a random variable with probability function $f(x)$.

Then the function $M_X(t)$ is called the moment generating function (MGF) of the random variable X and is defined

by

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x) ; \text{ if } X \text{ is discrete}$$
$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx ; \text{ if } X \text{ is continuous}$$

Problem:

A discrete random variable X has a probability function as shown in the following table:

Values of $X: x$	-3	-2	0	1	2
$P(X=x)=f(x)$	0.10	0.30	0.15	0.40	0.05

Find $E(X)$ and $V(X)$.

Solⁿ:

By definition we know

$$\mu = E(X) = \sum_{x=-3}^2 x f(x)$$

$$= (-3)(0.10) + (-2)(0.30) + (0)(0.15) + (1)(0.40) + (2)(0.05)$$

$$= -0.4$$

$$V(X) = E(X - \mu)^2 = \sum_{x=-3}^2 (X - \mu)^2 f(x)$$

$$= (-3 + 0.4)^2 (0.1) + (-2 + 0.4)^2 (0.30) + (0 + 0.4)^2 (0.15) + (1 + 0.4)^2 (0.40) + (2 + 0.4)^2 (0.05) = 2.54$$

Problem :

Using the formula $V(X) = E(X^2) - \mu^2$ find the variance & standard deviation (SD) of the previous probability function (see previous example)

Solⁿ :

By definition we know,

$$\begin{aligned} E(X^2) &= \sum x^2 f(x) \\ &= (-3)^2(0.10) + (-2)^2(0.30) + (0)^2(0.15) \\ &\quad + (1)^2(0.40) + (2)^2(0.05) \\ &= 2.7 \end{aligned}$$

Now,

$$\begin{aligned} V(X) &= E(X^2) - \mu^2 \\ &= 2.7 - (-0.4)^2 \\ &= 2.54 \quad \underline{\text{Ans.}} \end{aligned}$$

Hence,

$$\text{SD, } \sigma = \sqrt{V(X)}$$

$$= \sqrt{2.54}$$

$$= 1.59$$

(Ans)

Problem:

In a coin-tossing game, a man is promised to receive TK. 5 if he gets all heads or all tails when three coins are tossed and he pays off (loses) TK. 3 if either one or two heads appear. How much is he expected to gain in the long run?

Solⁿ:

The random variable X here is the amount of money (in Tk.) the man can win. Here, the r.v X will take on a value 5 when the coins show all heads ^{or all tails} and -3, otherwise

The table below shows the outcomes of the experiment, value of X and its associated probabilities:

Outcome :	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$X=x$	5	-3	-3	-3	-3	-3	-3	5
$P(X=x)=f(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

It appears from the above table that the variable X assumes values -3 and +5 with probabilities $\frac{6}{8}$ and $\frac{2}{8}$, respectively. Since the value -3 occurs 6 times and 5 occurs 2 times, the expected

value of X is

$$\begin{aligned} E(X) &= \sum_x x f(x) = -3\left(\frac{6}{8}\right) + 5\left(\frac{2}{8}\right) \\ &= -\frac{18}{8} + \frac{10}{8} \\ &= -1 \end{aligned}$$

Thus the man is expected to lose Tk. 1 in the long run.

Now let us examine what happens if the man receives Tk. 5 for all heads or all tails, Tk. 0 for 2 heads and pays off Tk. 3 for 1 head.

Now, the random variable X will take on values 5, 0 and -3 with associated probabilities $\frac{2}{8}$, $\frac{3}{8}$, and $\frac{3}{8}$, respectively.

The expected value in this case will be

$$\begin{aligned} E(X) &= \sum_x x f(x) \\ &= 5\left(\frac{2}{8}\right) + 0\left(\frac{3}{8}\right) + (-3)\left(\frac{3}{8}\right) \\ &= \frac{10}{8} + 0 - \frac{9}{8} \\ &= \frac{1}{8} = 0.125 \end{aligned}$$

This shows that the man will be marginally gainer winning only 12.5 paisa if the payment is made as designed above.

Problem:

A life insurance company in Bangladesh offers to sell a TK. 25000 one-year term life insurance policy to a 25 year-old man for a premium of TK. 2500. According to Bangladesh life table, the probability of surviving one year for a 25-years-old man is 0.97 and of his dying is 0.03. What is the company's expected gain in the long-run?

Solⁿ:

The gain X is a random variable that may take on the values TK. 2500, if the man survives or $2500 - 25000 = -\text{TK } 22500$ if he dies. Consequently, the probability distribution of X is as follows:

$X: x$	2500	-22500
$f(x)$	0.97	0.03

$$\therefore E[X] = 2500 * 0.97 - 22500 * 0.03$$

$$= 1750$$

Thus the company's ultimate gain is Tk. 1750.

Problem:

Let, X denotes the number of spots showing on the face of a well-balanced die after it is rolled once. If $Y = X^2 + 2X$, find

(a) $E[X]$, $E[Y]$, $E[X^2]$, $E[Y^2]$

(b) $V[X]$ and $V[Y]$.

Solⁿ:

The random variables X and Y together with their probability distributions are shown in the following table:

$X: x$	1	2	3	4	5	6
$Y: y$	3	8	15	24	35	48
$f(x) = f(y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E[X] = \sum_x x f(x) = \frac{1}{6} (1+2+3+4+5+6) = 3.5$$

$$\& E[Y] = \sum_y y f(y) = \frac{1}{6} (3+8+15+24+35+48) = 22.17$$

To find the variance of X , we find first $E[X^2]$

$$E[X^2] = \sum_x x^2 f(x)$$

$$= \frac{1}{6} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2)$$

$$\therefore E[X^2] = 15$$

Similarly,

$$E[Y^2] = \sum_y y^2 f(y)$$

$$= \frac{1}{6} (3^2 + 8^2 + 15^2 + 24^2 + 35^2 + 48^2)$$

$$\therefore E[Y^2] = 4403$$

Now,

$$\sigma_x^2 = E[X^2] - \{E[X]\}^2$$

$$= 15 - (3.5)^2$$

$$\therefore \sigma_x^2 = 2.75$$

Also,

$$\sigma_y^2 = E[Y^2] - \{E[Y]\}^2$$

$$= 4403 - (22.17)^2$$

$$\therefore \sigma_y^2 = 3911.49$$

Ans